Manual Disease Deep Dive Template

Disease / Project

Disease name:
`[fill in disease name]`

Date started:
`[fill in date]`

Main question:
`What am I trying to understand about this disease?`

Example:

```
How does this disease arise biologically, what genes/proteins are involved, and how would I manua
```

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1. Quick Overview

Plain-English Disease Summary

What is the disease?

```
[Write 3-5 sentences explaining the disease in simple language.]
```

Who does it affect?

```
[Age of onset, population, inheritance, prevalence if known.]
```

Main clinical features / symptoms

- `[symptom 1]`
- `[symptom 2]`
- `[symptom 3]`

Why is this disease biologically interesting?

```
[Explain why this disease matters: rare disease, genetic cause, protein dysfunction, pathway rele
```

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2. Source Map

Use this table to track where each type of information comes from.

Information Needed	Main Source	Link Used	Notes
Disease overview	Orphanet	`[link]`	`[notes]`
Human genetics	OMIM	`[link]`	`[notes]`
Disease terminology	MedGen / MONDO	`[link]`	`[notes]`
Gene information	NCBI Gene / Ensembl / HGNC	`[link]`	`[notes]`
Protein information	UniProt	`[link]`	`[notes]`
Protein domains	InterPro / Pfam	`[link]`	`[notes]`
Protein structure	AlphaFold / PDB	`[link]`	`[notes]`
Variants	ClinVar / gnomAD	`[link]`	`[notes]`

Information Needed	Main Source	Link Used	Notes
Pathways	Reactome / KEGG	`[link]`	`[notes]`
Literature	PubMed / Europe PMC / Google Scholar	`[link]`	`[notes]`

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3. Disease Database Deep Dive

3.1 Orphanet

Search query

`[disease name] site:orpha.net`

Link used

`[Paste Orphanet link here]`

Orpha ID

`[fill in]`

Disease synonyms

- `[synonym 1]`
- `[synonym 2]`
- `[synonym 3]`

Disease classification

`[Where does Orphanet classify this disease? Neurological? Metabolic? Developmental? etc.]`

Prevalence

`[fill in]`

Inheritance pattern

`[autosomal dominant / autosomal recessive / X-linked / mitochondrial / unknown]`

Associated genes listed by Orphanet

Gene	Evidence / Notes
`[gene]`	`[notes]`
`[gene]`	`[notes]`

My understanding

`[In your own words, explain what Orphanet is telling you.]`

3.2 OMIM

Search query

`[disease name] OMIM`

Link used

`[Paste OMIM link here]`

OMIM disease ID

`[fill in]`

OMIM phenotype title

`[fill in]`

Known gene(s)

Gene	OMIM Gene ID	Notes
[gene]	[ID]	[notes]

Inheritance

`[fill in]`

Key OMIM points

- [point 1]
- [point 2]
- [point 3]

What OMIM adds that Orphanet did not

`[Write what extra genetic/clinical detail OMIM provides.]`

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4. Gene Deep Dive

Repeat this section for each important gene.

Gene 1

Gene symbol

`[fill in]`

Full gene name

`[fill in]`

Main links

Database	Link
NCBI Gene	[link]
Ensembl	[link]
HGNC	[link]
GeneCards	[link]

4.1 Gene Identity

Chromosome location

`[fill in]`

Aliases / alternative names

- `[alias 1]`
- `[alias 2]`
- `[alias 3]`

What does this gene do?

`[Plain-English summary.]`

Where is it expressed?

`[Tissues/cells/organs if known.]`

Why is this gene relevant to the disease?

`[Explain the causal relationship if known.]`

4.2 Gene-to-Disease Evidence

Evidence Type	What I Found	Source Link
Disease association	[fill in]	[link]
Inheritance	[fill in]	[link]
Known pathogenic variants	[fill in]	[link]
Functional evidence	[fill in]	[link]
Animal/cell model evidence	[fill in]	[link]

Confidence level

`High / Medium / Low`

Why?

`[Explain why you trust or do not trust the gene-disease link.]`

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5. Variant / Mutation Deep Dive

Use this section if the disease involves known variants.

Main variant sources

Database	Link Used	Notes
ClinVar	[link]	[notes]
gnomAD	[link]	[notes]
dbSNP	[link]	[notes]
LOVD	[link]	[notes]

Variant Table

Variant	Gene	Variant Type	Clinical Significance	Frequency	Source
[variant]	[gene]	[missense / nonsense / frameshift / splice / deletion]	[pathogenic / likely pathogenic / VUS / benign]	[frequency]	[link]

Variant Interpretation

Most important variant(s)

[fill in]

What does the mutation change?

[Example: amino acid substitution, premature stop codon, altered splicing, loss of function.]

Why might this cause disease?

[Explain the biological mechanism.]

Is this variant common in healthy populations?

[Use gnomAD if available.]

My confidence

High / Medium / Low

Reason

[fill in]

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6. Protein Deep Dive

Protein Identity

Gene

[fill in]

Protein name

[fill in]

UniProt link

[Paste UniProt link here]

UniProt accession ID

[fill in]

Protein Function

What does this protein normally do?

[Plain-English explanation.]

Where is the protein located in the cell?

[cell membrane / nucleus / mitochondria / cytoplasm / extracellular / other]

What biological process is it involved in?

[fill in]

What happens if this protein fails?

[Explain disease mechanism.]

Protein Domains

Domain	Position	Source	Why It Matters
[domain]	[amino acid range]	[InterPro/Pfam/UniProt link]	[notes]

Important domain links

Database	Link
InterPro	[link]
Pfam	[link]

My understanding of the protein

[Explain this protein as if teaching it to someone else.]

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7. Protein Structure Deep Dive

Structure Sources

Source	Link	Notes
AlphaFold	[link]	[notes]
PDB	[link]	[notes]

AlphaFold / PDB Information

Structure available?

Yes / No

Structure type

Predicted / Experimental / Both

Important structural regions

[Folded domains, disordered regions, binding pockets, transmembrane regions, active sites.]

Are disease variants located in important regions?

[Explain if known.]

Structural Interpretation

Could the mutation disrupt protein folding?

Yes / No / Unsure

Could the mutation affect a binding site or functional domain?

Yes / No / Unsure

Could the mutation affect protein stability?

Yes / No / Unsure

Explanation

[Write your reasoning.]

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8. Pathway / Network Deep Dive

Pathway Sources

Database	Link	Notes
Reactome	[link]	[notes]
KEGG	[link]	[notes]

Database	Link	Notes
STRING	[link]	[notes]

Pathways Involved

Pathway	Source	Why It Matters
[pathway]	[link]	[notes]

Interaction Partners

Interacting Protein / Gene	Source	Possible Relevance
[protein/gene]	[STRING/UniProt/etc.]	[notes]

Systems-Level Interpretation

What larger biological system is affected?

`[Example: ion transport, mitochondrial function, DNA repair, protein degradation, immune signalli`

What downstream effects might happen?

`[fill in]`

My understanding

`[Explain how the gene/protein fits into the bigger biological picture.]`

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9. Literature Validation

PubMed Search Queries

Use several versions.

`[disease name] [gene name]`

`[disease name] [gene name] mutation`

`[gene name] protein function disease`

`[disease name] genotype phenotype`

`[disease name] therapy review`

Key Papers

Paper Title	Year	Link	What It Shows	Evidence Strength
[title]	[year]	[link]	[summary]	High / Medium / Low

Evidence Summary

Strongest evidence found

[fill in]

Any conflicting evidence?

[fill in]

Most useful review paper

[link + notes]

Most useful experimental paper

[link + notes]

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10. Therapeutic / Translational Relevance

Existing Treatments

Treatment	Mechanism	Source	Notes
[treatment]	[how it works]	[link]	[notes]

Experimental / Emerging Therapies

Therapy Type	Example	Source	Notes
Gene therapy	[fill in]	[link]	[notes]
Small molecule	[fill in]	[link]	[notes]
Protein replacement	[fill in]	[link]	[notes]
RNA therapy	[fill in]	[link]	[notes]

Therapeutic Interpretation

What is the main therapeutic challenge?

[fill in]

What kind of therapy would make biological sense?

```
[gene replacement / enzyme replacement / small molecule / RNA therapy / symptomatic treatment / u
```

Why?

```
[Explain the reasoning.]
```

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11. Computational Workflow Equivalent

This is the section where you connect the manual work to what the AI/code is doing.

11.1 Disease Search

Manual version

```
Search Orphanet / OMIM / MedGen for disease name.
```

Code-like version

```
disease_name = "[fill in disease name]"

# Goal:
# 1. Search disease database
# 2. Retrieve disease identifiers
# 3. Extract linked genes
# 4. Store source links
```

Expected output

```
Disease ID:
Synonyms:
Associated genes:
Source links:
```

11.2 Gene Lookup

Manual version

```
Search NCBI Gene / Ensembl / HGNC for gene symbol.
```

Code-like version

```
gene_symbol = "[fill in gene]"

# Goal:
# 1. Resolve official gene symbol
# 2. Collect aliases
# 3. Get chromosome location
# 4. Link gene to disease evidence
```

Expected output

```
Gene symbol:
Full name:
Aliases:
Location:
Disease evidence:
```

11.3 UniProt Protein Lookup

Manual version

Search UniProt for the gene/protein.

Example code

```python
import requests

gene = "[GENE_SYMBOL]"

url = "https://rest.uniprot.org/uniprotkb/search"
params = {
    "query": f"gene_exact:{gene} AND organism_id:9606",
    "format": "json"
}

response = requests.get(url, params=params)
data = response.json()

print(data.keys())
```

What I should extract

```
Protein name:
UniProt accession:
Function:
Subcellular location:
Domains:
Sequence:
```

11.4 PubMed Literature Search

Manual version

Search PubMed using disease + gene + mutation terms.

Example code

```python
from Bio import Entrez

Entrez.email = "your_email@example.com"

query = "[DISEASE_NAME] [GENE_SYMBOL] mutation"
```

```python
handle = Entrez.esearch(
    db="pubmed",
    term=query,
    retmax=10
)

record = Entrez.read(handle)

print(record["IdList"])
```

What I should extract

```
Top papers:
Review papers:
Experimental evidence:
Clinical evidence:
```

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12. AI Output Validation Checklist

Use this whenever AI gives you an answer.

Disease Claims

- ☐ Did AI provide a disease ID?
- ☐ Did I verify it in Orphanet or OMIM?
- ☐ Did AI list synonyms?
- ☐ Did I check whether synonyms are real?

Gene Claims

- ☐ Did AI identify associated genes?
- ☐ Did I verify each gene in NCBI / OMIM / Orphanet?
- ☐ Did AI confuse aliases with official symbols?
- ☐ Did I check HGNC for official names?

Variant Claims

- ☐ Did AI name specific variants?
- ☐ Did I verify variants in ClinVar?
- ☐ Did I check whether variants are pathogenic or uncertain?
- ☐ Did I check population frequency in gnomAD?

Protein Claims

- ☐ Did AI identify the correct protein?
- ☐ Did I verify it in UniProt?
- ☐ Did I check domains?
- ☐ Did I check cellular location?

Structure Claims

- ☐ Did AI mention AlphaFold or PDB?
- ☐ Did I verify structure availability?
- ☐ Did I check whether mutation positions actually map to important regions?

Literature Claims

- ☐ Did AI cite papers?
- ☐ Did I open the papers or abstracts?
- ☐ Did I check dates?
- ☐ Did I check whether evidence is review, clinical, or experimental?

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13. Final Disease Summary

Disease

`[fill in]`

Main gene(s)

`[fill in]`

Main protein(s)

`[fill in]`

Main biological mechanism

`[Explain how the disease happens biologically.]`

Main variants / mutations

`[fill in]`

Main affected pathway

`[fill in]`

Strongest evidence

`[fill in]`

Weakest / uncertain evidence

`[fill in]`

Therapeutic relevance

`[fill in]`

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14. My Understanding In Plain English

Write this last.

```
This disease is caused by...

The main gene involved is...

That gene produces...

The protein normally does...

When the gene/protein is disrupted...

This leads to disease because...

The strongest evidence comes from...

The main uncertainty is...
```

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15. Things I Still Don't Understand

Use this as your learning list.

Question	Why It Matters	Who/What Can Help
[question]	[notes]	[AI / paper / database / supervisor]
[question]	[notes]	[AI / paper / database / supervisor]

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16. Glossary Terms To Add

Add these to `docs/GLOSSARY.md`.

Term	My Definition	Source
[term]	[plain-English definition]	[link]
[term]	[plain-English definition]	[link]

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17. Final Confidence Score

How well do I understand this disease?

```
1 / 5 = I barely understand it
2 / 5 = I understand the basic disease
3 / 5 = I understand disease → gene → protein
```

```
4 / 5 = I can explain mechanism and evidence
5 / 5 = I can independently validate AI output
```

Score

[fill in]

Why?

[fill in]

Next action

[What is the next thing I should verify or learn?]